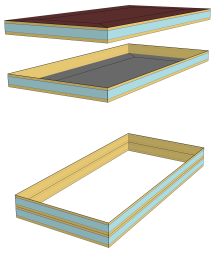
SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Commercial-Institutional
Building type	Office
Building subtype	High-rise (10+ storeys)
Vintage	After 1983

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation.

BUILDING CHARACTERISTICS

Total floor area [m²]	49,583
Conditioned floor area [m²]	49,583
Above-ground storeys	16
Climate zone	6A
Weather file	Montreal-Trudeau Intl AP QC CAN CWEC2016 WMO#=716270

WINDOW-TO-WALL RATIO [%]

Average	60.0
North	60.0
East	60.0
South	60.0
West	60.0

HVAC SYSTEM

Heating	Electric boiler (hot water) + Electric baseboards
Cooling	Water-cooled chiller
Air distribution	VAV with reheat (1 air loop)
Heat recovery	None

ENVELOPE & INTERNAL LOADS

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²K]	0.44
Roof [W/m²K]	0.21
Exposed floors / slabs [W/m²K]	2.95
Infiltration [m³/h/m² ext. surface]	0.82

GLAZING

Glazing U-factor [W/m²K]	1.30
SHGC	0.35

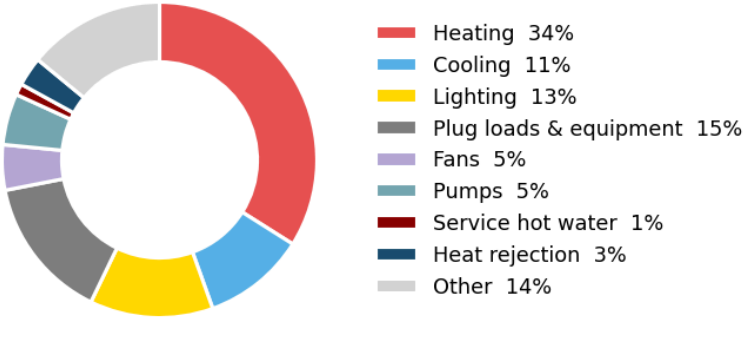
INTERNAL LOADS (MODEL INPUTS)

Occupant density [m²/person]	25.0
Lighting power density [W/m²]	10.8
Equipment power density [W/m²]	8.5

ANNUAL END USES [kWh/year]

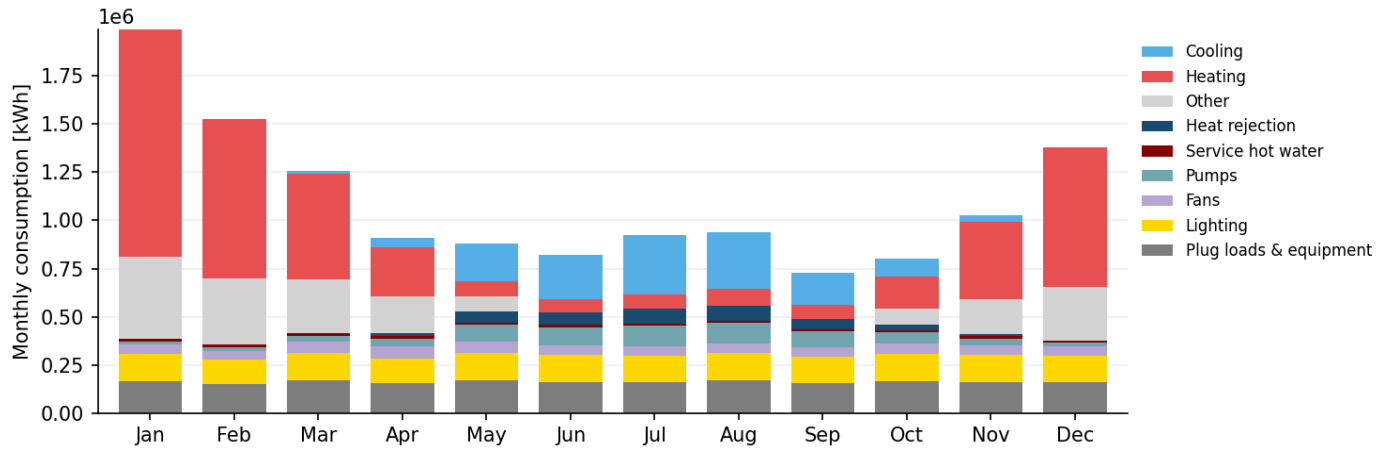
END USE	ELECTRICITY	TOTAL	kWh/m²
Heating	4,474,845	4,474,845	90.2
Cooling	1,393,570	1,393,570	28.1
Interior lighting	1,669,222	1,669,222	33.7
Plug loads & equipment	1,942,681	1,942,681	39.2
Fans	611,867	611,867	12.3
Pumps	692,161	692,161	14.0
Heat rejection	394,644	394,644	8.0
Humidification	1,851,808	1,851,808	37.3
Service hot water	152,008	152,008	3.1
Total		**13,182,801**	**265.9**

SHARE OF ANNUAL CONSUMPTION



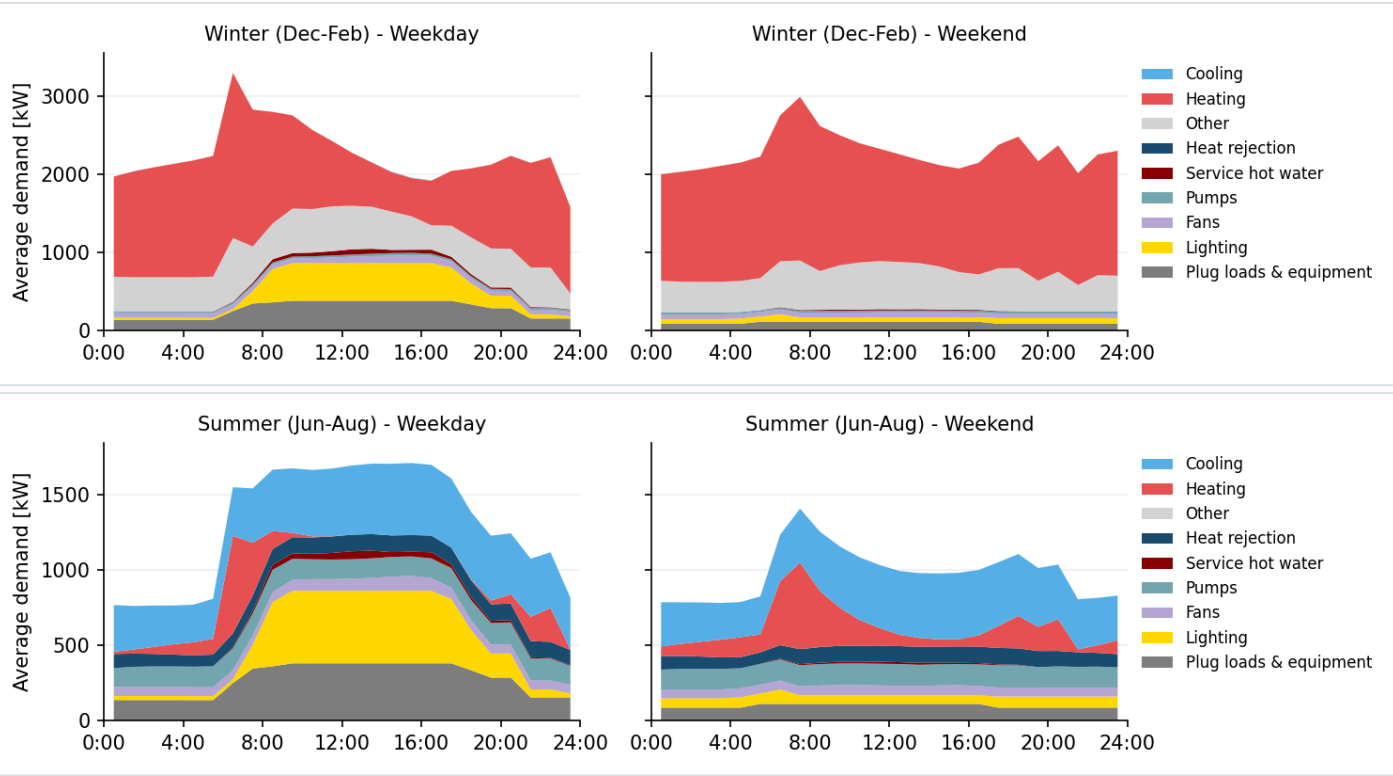
ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS	PEAK ELECTRIC DEMAND
266 kWh/m²/yr (total area)	13,182,801 kWh/yr	13,182,807 kWh/yr	0 kWh/yr	5,840 kW · 117.8 W/m²

MONTHLY CONSUMPTION BY END USE



Monthly energy consumption by end use, all fuels combined.

TYPICAL DAY PROFILES — WEEKDAY VS. WEEKEND



Average hourly demand by end use for winter and summer. Weekdays (Monday to Friday) are shown on the left and weekends and holidays on the right, showing the office occupancy schedule and HVAC set-back.

NOTES & ASSUMPTIONS

- Archetype energy model simulated with OpenStudio/EnergyPlus. Values are model outputs for the reference weather year and are not measured data.
- Energy use intensity is normalized by the total floor area (49,583 m²). Offices are not normalized per unit.
- Archetype: High-rise office (10+ storeys), built after 1983.
- HVAC entries marked "(typical)" are not modelled explicitly and come from the archetype assumption table.
- Peak electric demand occurs on 29-JAN-07:20.